HW 6

1. Let W be a subspace of an inner product space V

a) Show $W^\perp \subseteq V$

contains $\vec{0}$: for $w \in W$, $\langle 0, w\rangle = 0$ by definition $\Rightarrow 0 \in W^\perp$

closed under $(+)$: let $v_1, v_2 \in W^\perp$. Then, we must show that $v_1 + v_2 \in W^\perp$, or $\langle v_1 + v_2, w\rangle = 0$

for $w \in W$, $\langle v_1 + v_2, w\rangle = \langle v_1, w\rangle + \langle v_2, w\rangle$ but since v_1 and v_2 are orthogonal to every vector in W, then $\langle v_1 + v_2, w\rangle = 0$, which means $v_1 + v_2 \in W^\perp$

closed under $(\cdot)$: let $v \in W^\perp$ and $c \in \mathbb{R}$. Then, we must show that $cv \in W^\perp$, or $\langle cv, w\rangle = 0$

for $w \in W$, $\langle cv, w\rangle = c\langle v, w\rangle$. Since $v \in W^\perp$, $\langle v, w\rangle = 0$, so $c\langle v, w\rangle = c \cdot 0 = 0$ and $cv \in W^\perp$

Because $W^\perp$ satisfies these 3 conditions, it is a subspace of V.

b) Show that $W \cap W^\perp = \{0\}$

Let $v \in W$ and $v \in W^\perp$. Then, for v to be in both W and $W^\perp$, then it must be orthogonal to itself, such that $\langle v, v\rangle = 0$ and orthogonal to all of $w_1 \in W$ and $w_2 \in W^\perp$ so that $\langle v, w_1\rangle = \langle v, w_2\rangle = 0$. The only vector that satisfies this is the zero vector, so $W \cap W^\perp = \{0\}$.

2. Suppose nonzero vectors $v_1, \dots, v_n \in \mathbb{R}^m$ are mutually orthogonal. Prove linear independence.

In order for $v_1, \dots, v_n$ to be mutually orthogonal, each vector v_i must be in a separate dimension.

$a_1, \dots, a_n \in \mathbb{R}$

for any $v \in \mathbb{R}^n$, $v = a_1 v_1 + \dots + a_n v_n = 0$, then if we choose an vector v_i and multiply both sides,

$$v_i^T (a_1 v_1 + \dots + a_n v_n) = v_i^T (0) \Rightarrow a_1 v_i^T v_1 + \dots + a_n v_i^T v_n = 0$$

But, because every vector is mutually orthogonal, i.e. $v_i^T v_j = 0$ for $i \neq j$, then every term dissapears other than the i-th term, so

$$a_i v_i^T v_i = 0 \Rightarrow a_i \|v_i\|_2^2 = 0$$

Since v_i must be positive, then $a_i = 0$, which shows linear independence.

3.

$$A = \begin{pmatrix} 1 & 0 & 1 \\ 0 & -2 & 0 \\ 1 & -2 & 2 \end{pmatrix} \qquad A = QR$$

$$u_1 = v_1 \Rightarrow q_1 = \frac{v_1}{\|v_1\|_2} = \frac{1}{\sqrt{1+1}}\begin{bmatrix}1\\0\\1\end{bmatrix} = \begin{bmatrix}1/\sqrt{2}\\0\\1/\sqrt{2}\end{bmatrix}$$

$$u_2 = v_2 - (v_2^T q_1)q_1 = \begin{bmatrix}0\\-2\\-2\end{bmatrix} - (-\sqrt{2})\begin{bmatrix}1/\sqrt{2}\\0\\1/\sqrt{2}\end{bmatrix} = \begin{bmatrix}0\\-2\\-2\end{bmatrix} + \begin{bmatrix}2/2\\0\\2/2\end{bmatrix} = \begin{bmatrix}1\\-2\\-1\end{bmatrix} \Rightarrow q_2 = \frac{1}{\sqrt{1+4+1}}\begin{bmatrix}1\\-2\\-1\end{bmatrix} = \begin{bmatrix}1/\sqrt{6}\\-2/\sqrt{6}\\-1/\sqrt{6}\end{bmatrix}$$

$$u_3 = v_3 - (v_3^T q_1)q_1 - (v_3^T q_2)q_2 = \begin{bmatrix}1\\0\\2\end{bmatrix} - \underbrace{\left(\frac{1}{\sqrt{2}} + \frac{2}{\sqrt{2}}\right)}_{3/\sqrt{2}}\begin{bmatrix}1/\sqrt{2}\\0\\1/\sqrt{2}\end{bmatrix} - \underbrace{\left(\frac{1}{\sqrt{6}} - \frac{2}{\sqrt{6}}\right)}_{-1/\sqrt{6}}\begin{bmatrix}1/\sqrt{6}\\-2/\sqrt{6}\\-1/\sqrt{6}\end{bmatrix} = \begin{bmatrix}1\\0\\2\end{bmatrix} - \begin{bmatrix}3/2\\0\\3/2\end{bmatrix} - \begin{bmatrix}-1/6\\1/3\\1/6\end{bmatrix}$$

$$= \begin{bmatrix}\frac{6}{6} - \frac{9}{6} + \frac{1}{6}\\ -1/3 \\ \frac{12}{6} - \frac{9}{6} - \frac{1}{6}\end{bmatrix} = \begin{bmatrix}-1/3\\-1/3\\1/3\end{bmatrix} \Rightarrow q_3 = \frac{1}{\underbrace{\sqrt{\frac{1}{9}+\frac{1}{9}+\frac{1}{9}}}_{1/\sqrt{3}}}\begin{bmatrix}-1/3\\-1/3\\1/3\end{bmatrix} = \begin{bmatrix}-\sqrt{3}/3\\-\sqrt{3}/3\\\sqrt{3}/3\end{bmatrix} = \begin{bmatrix}-1/\sqrt{3}\\-1/\sqrt{3}\\1/\sqrt{3}\end{bmatrix}$$

$$\boxed{A = QR = \underbrace{\begin{bmatrix}1/\sqrt{2} & 1/\sqrt{6} & -1/\sqrt{3}\\ 0 & -2/\sqrt{6} & -1/\sqrt{3}\\ 1/\sqrt{2} & -1/\sqrt{6} & 1/\sqrt{3}\end{bmatrix}}_{Q}\underbrace{\begin{bmatrix}\sqrt{2} & -\sqrt{2} & 3/\sqrt{2}\\ 0 & \sqrt{6} & -1/\sqrt{6}\\ 0 & 0 & 1/\sqrt{3}\end{bmatrix}}_{R}}$$

$$b = \begin{bmatrix}0\\-1\\-2\end{bmatrix}$$

$$y = Q^T b = \begin{bmatrix}1/\sqrt{2} & 0 & 1/\sqrt{2}\\ 1/\sqrt{6} & -2/\sqrt{6} & -1/\sqrt{6}\\ -1/\sqrt{3} & -1/\sqrt{3} & 1/\sqrt{3}\end{bmatrix}\begin{bmatrix}0\\-1\\-2\end{bmatrix} = \begin{bmatrix}y_1\\y_2\\y_3\end{bmatrix} \Rightarrow \begin{bmatrix}-2/\sqrt{2}\\ 2/\sqrt{6} + 2/\sqrt{6}\\ 1/\sqrt{3} - 2/\sqrt{3}\end{bmatrix} = \begin{bmatrix}-\sqrt{2}\\4/\sqrt{6}\\-1/\sqrt{3}\end{bmatrix}$$

$$Rx = y \Rightarrow \begin{bmatrix}\sqrt{2} & -\sqrt{2} & 3/\sqrt{2}\\ 0 & \sqrt{6} & -1/\sqrt{6}\\ 0 & 0 & 1/\sqrt{3}\end{bmatrix}\begin{bmatrix}x_1\\x_2\\x_3\end{bmatrix} = \begin{bmatrix}-\sqrt{2}\\4/\sqrt{6}\\-1/\sqrt{3}\end{bmatrix} \Rightarrow$$

$x_3 = -1$

$\sqrt{6}x_2 + \frac{1}{\sqrt{6}} = 4/\sqrt{6} \Rightarrow \sqrt{6}x_2 = 3/\sqrt{6} \Rightarrow x_2 = \frac{1}{2}$

$\sqrt{2}x_1 - \sqrt{2}\left(\frac{1}{2}\right) + 3/\sqrt{2}(-1) = -\sqrt{2} \Rightarrow x_1 - \frac{1}{2} - \frac{3}{2} = -1$

$\Rightarrow x_1 = -1 + \frac{1}{2} + \frac{3}{2} \Rightarrow x_1 = 1$

$$\boxed{x = \left(1 \;\; 1/2 \;\; -1\right)^T}$$

4. Let $A \in \mathbb{R}^{m \times n}$ be full rank, $m \geq n$

a) Show that $A = QR$ is not unique if sign is not specified on diagonal of R

Let $A = QR = Q'R'$ be two factorizations for A. Then, we can construct a diagonal matrix $S = \text{diag}(s_1, \dots, s_n)$ where $s_i \in \{1, -1\}$ and define $Q' = QS$ and $R' = SR$. Then, it follows that

$$A = Q'R' = (QS)(SR) = Q(SS)R = QS^2R = QIR = QR = A$$

equals the identity
b/c S is diag of 1's

Furthermore, we must verify that Q' is still orthogonal, or that $(Q')^TQ' = I$

$$(Q')^TQ' = (QS)^TQS = S^TQ^TQS = S^T(I)S = S^TS = I$$

each q_i is orthonormal
$\Rightarrow Q$ is orthogonal

S is orthonormal
b/c it is diag of 1's

We can also verify that R' is still upper triangular because multiplying with a diagonal matrix does not affect the zero structure of R. Thus, we have proven there are multiple ways to form a QR factorization with non-unique signs in the diagonal of R.

b) If $m > n$, then QR must be extended such that Q is $m \times m$, like

$$\begin{array}{c} \\ m \end{array}\!\!\overset{n}{\begin{bmatrix} A \end{bmatrix}} = m\overset{m}{\begin{bmatrix} \quad Q \quad \end{bmatrix}}\overset{n}{\begin{bmatrix} R \end{bmatrix}}m = \begin{bmatrix} Q_1 & Q_2 \end{bmatrix}\begin{bmatrix} R \\ 0 \end{bmatrix}$$

But, in order to extend Q, we must use an orthonormal basis for Q_2.
Because the dimension ≥ 2, there are infinite choices for an orthonormal basis.
Thus, the full QR factorization can never be unique.

5. a) Let $q_1, \dots, q_i \in \mathbb{R}^n$.

$$v^T q_i = \left(v^T - \left(\sum_{j=1}^{i-1} (v^T q_j) q_j\right)^T\right) q_i = \left(v^T - \sum_{j=1}^{i-1} (v^T q_j) q_j^T\right) q_i = v^T q_i - \sum_{j=1}^{i-1} (v^T q_j)(q_j^T q_i)$$

$\langle v, q_j \rangle$ is a scalar $\Rightarrow$ transpose does not affect it

this is 0 for all $j \neq i$

$= v^T q_i - 0 = v^T q_i$ $\Rightarrow$ thus, we have proven this equality.

d) $A = \begin{pmatrix} 1 & 1 & 1 \\ \varepsilon & 0 & 0 \\ 0 & \varepsilon & 0 \\ 0 & 0 & \varepsilon \end{pmatrix}$

classical GS

$$q_1 = \frac{v_1}{\|v_1\|} = \frac{1}{\sqrt{1+\varepsilon^2}} \begin{bmatrix} 1 \\ \varepsilon \\ 0 \\ 0 \end{bmatrix} \approx \begin{bmatrix} 1 \\ \varepsilon \\ 0 \\ 0 \end{bmatrix}$$

$$u_2 = v_2 - \langle v_2, q_1 \rangle q_1 = \begin{bmatrix} 1 \\ 0 \\ \varepsilon \\ 0 \end{bmatrix} - \begin{bmatrix} 1 \\ \varepsilon \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ -\varepsilon \\ \varepsilon \\ 0 \end{bmatrix} \quad q_2 = \frac{1}{\sqrt{2}\,\varepsilon} \begin{bmatrix} 0 \\ -\varepsilon \\ \varepsilon \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ -1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{bmatrix}$$

$$u_3 = v_3 - \langle v_3, q_1 \rangle q_1 - \langle v_3, q_2 \rangle q_2 = \begin{bmatrix} 0 \\ -\varepsilon \\ 0 \\ \varepsilon \end{bmatrix} \quad q_3 = \frac{1}{\sqrt{2}\,\varepsilon} \begin{bmatrix} 0 \\ -\varepsilon \\ 0 \\ \varepsilon \end{bmatrix} = \begin{bmatrix} 0 \\ -1/\sqrt{2} \\ 0 \\ 1/\sqrt{2} \end{bmatrix}$$

Inner product! $q_2^T q_3 = \begin{bmatrix} 0 & -1/\sqrt{2} & 1/\sqrt{2} & 0 \end{bmatrix} \begin{bmatrix} 0 \\ -1/\sqrt{2} \\ 0 \\ 1/\sqrt{2} \end{bmatrix} = \frac{1}{2}$, far from orthogonal

Modified GS

$$v_3 = \begin{bmatrix} 0 \\ -\varepsilon \\ 0 \\ \varepsilon \end{bmatrix}$$

$$r_{23} = q_2^T v_3 = \begin{bmatrix} 0 & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \end{bmatrix} \begin{bmatrix} 0 \\ -\varepsilon \\ 0 \\ \varepsilon \end{bmatrix} = \frac{\varepsilon}{\sqrt{2}}$$

$v_3 = v_3 - r_{23} q_2$ ... $q_2^T q_3 \approx 0$, after normalization

```python
import numpy as np


def classical_qr(A):
    m, n = A.shape
    Q = np.zeros((m, n))
    R = np.zeros((n, n))

    for i in range(n):
        v = A[:, i].copy()

        for j in range(i):
            R[j, i] = Q[:, j] @ A[:, i]
            v -= R[j, i] * Q[:, j]

        R[i, i] = np.linalg.norm(v)
        Q[:, i] = v / R[i, i]

    return Q, R


def modified_qr(A):
    m, n = A.shape
    Q = np.zeros((m, n))
    R = np.zeros((n, n))
    V = A.copy()

    for i in range(n):
        R[i, i] = np.linalg.norm(V[:, i])
        Q[:, i] = V[:, i] / R[i, i]

        for j in range(i + 1, n):
            R[i, j] = Q[:, i] @ V[:, j]
            V[:, j] = V[:, j] - R[i, j] * Q[:, i]

    return Q, R


def qr_errors(A, Q, R):
    residual = np.linalg.norm(A - Q @ R, 2)
    orthogonality = np.linalg.norm(Q.T @ Q - np.eye(Q.shape[1]), 2)
    return residual, orthogonality


eps = np.finfo(float).eps

A = np.array([
    [1, 1, 1],
    [eps, 0, 0],
    [0, eps, 0],
    [0, 0, eps]
], dtype=float)

Q_cgs, R_cgs = classical_qr(A)
Q_mgs, R_mgs = modified_qr(A)

res_cgs, ortho_cgs = qr_errors(A, Q_cgs, R_cgs)
res_mgs, ortho_mgs = qr_errors(A, Q_mgs, R_mgs)

print("CGS residual:", res_cgs)
print("CGS orthogonality:", ortho_cgs)

print("MGS residual:", res_mgs)
print("MGS orthogonality:", ortho_mgs)

''' TERMINAL OUTPUT
CGS residual: 6.372400086712262e-33
CGS orthogonality: 0.5000000000000001
MGS residual: 6.0612148999478584e-33
MGS orthogonality: 3.083989583803855e-16
'''
```